

1. Velocity and constant acceleration
a. Displacement and velocity

WORKED EXAMPLE 1.1

A car travels 9 km in 15 minutes at constant speed. Find its speed in m s^{-1} .

Answer

$$\begin{aligned} 9 \text{ km} &= 9000 \text{ m} \text{ and} \\ 15 \text{ minutes} &= 900 \text{ s} \end{aligned}$$

Convert to units required for the answer, which are SI units.

$$\begin{aligned} s &= vt \\ \text{so } 9000 &= 900v \\ v &= 10 \text{ m s}^{-1} \end{aligned}$$

Substitute into the equation for displacement and solve.

WORKED EXAMPLE 1.2

A cyclist travels at 5 m s^{-1} for 30 s then turns back, travelling at 3 m s^{-1} for 10 s. Find her displacement in the original direction of motion from her starting position.

Answer

$$\begin{aligned} s &= vt \\ \text{So } s_1 &= 5 \times 30 \\ &= 150 \end{aligned}$$

Separate the two stages of the journey.

$$\begin{aligned} \text{and } s_2 &= -3 \times 10 \\ &= -30 \end{aligned}$$

Remember travelling back means a negative velocity and a negative displacement.

$$\begin{aligned} \text{So the total displacement is} \\ s &= 150 + (-30) \\ &= 120 \text{ m} \end{aligned}$$

b. Acceleration

WORKED EXAMPLE 1.4

A parachutist falls from rest to 49 m s^{-1} over 5 s. Find her acceleration.

Answer

$$\begin{aligned} a &= \frac{v - u}{t} \\ &= \frac{49 - 0}{5} \\ &= 9.8 \text{ m s}^{-2} \end{aligned}$$

Make sure you use the correct units, which are m s^{-2} .



TIP

'Rest' means not moving, so velocity is zero.

WORKED EXAMPLE 1.5

A tractor accelerates from 5 m s^{-1} to 9 m s^{-1} at 0.5 m s^{-2} . Find the distance covered by the tractor over this time.

Answer

$$a = \frac{v - u}{t}$$

Substitute into $a = \frac{v - u}{t}$ first to find t .

$$\text{So } 0.5 = \frac{9 - 5}{t}$$

$$0.5t = 4$$

$$t = 8 \text{ s}$$

$$s = \frac{1}{2}(u + v)t$$

Substitute into $s = \frac{1}{2}(u + v)t$ to find s .

$$= \frac{1}{2}(5 + 9) \times 8$$

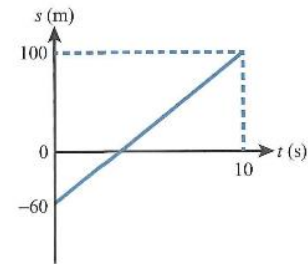
$$= 56 \text{ m}$$

c. Equations of constant acceleration

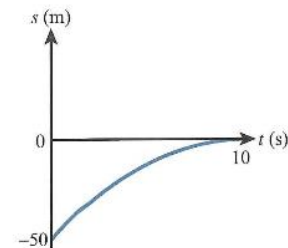
- 3 Assuming constant acceleration, find v when $s = 6 \text{ m}$, $u = 5 \text{ m s}^{-1}$ and $a = -2 \text{ m s}^{-2}$ if the object has changed direction during the motion.
- 4 Assuming constant acceleration, find u when $s = 60 \text{ m}$, $v = 13 \text{ m s}^{-1}$ and $a = 1 \text{ m s}^{-2}$ if the object has not changed direction during the motion.
- 5 a Assuming constant acceleration, find v when $s = 18 \text{ m}$, $u = 3 \text{ m s}^{-1}$ and $a = 2 \text{ m s}^{-2}$.
b Why is it not necessary to specify in this question whether the object has changed direction during the motion?
- 6 A car is travelling at a velocity of 20 m s^{-1} when the driver sees the traffic lights ahead change to red. He decelerates at a constant rate of 4 m s^{-2} and comes to a stop at the lights. Find how far away from the lights the driver started braking.
- 7 An aeroplane accelerates at a constant rate along a runway from rest until taking off at a velocity of 60 m s^{-1} . The runway is 400 m long. Find the acceleration of the aeroplane.
- 8 An aeroplane accelerates from rest along a runway at a constant rate of 4 m s^{-2} . It needs to reach a velocity of 80 m s^{-1} to take off. Find how long the runway needs to be.
- 9 A motorcyclist sees that the traffic lights are red 40 m ahead of her. She is travelling at a velocity of 20 m s^{-1} and comes to rest at the lights. Find the deceleration she experiences, assuming it is constant.
- M** 10 A driver sees the traffic lights change to red 240 m away when he is travelling at a velocity of 30 m s^{-1} . To avoid wasting fuel, he does not brake, but lets the car slow down naturally. The traffic lights change to green after 12 s , at the same time as the driver arrives at the lights.
a Find the speed at which the driver goes past the lights.
b What assumptions have been made to answer the question?

d. Displacement-time graph

- 3 Sketch the displacement–time graphs from the information given. In each case consider forwards to be the positive direction and the traffic lights to be the point from which displacement is measured. Remember to include the values for time and displacement at any points where the motion changes.
- A car is waiting at rest at the traffic lights. It accelerates at a constant rate of 3 m s^{-2} for 5 s, then remains at constant speed for the next 10 s.
 - A motorbike passes the traffic lights at a constant speed of 10 m s^{-1} . After 6 s it starts to slow at a constant rate of 2 m s^{-2} until it comes to rest.
 - A truck is moving at a constant speed of 8 m s^{-1} and is approaching the traffic lights 60 m away. When it is 20 m away it accelerates at a constant rate of 2 m s^{-2} to get past the lights before they change colour.
 - A scooter accelerates from rest 100 m before the traffic lights at a constant rate of 1.5 m s^{-2} until it reaches 6 m s^{-1} . It then travels at this speed until it reaches a point 50 m beyond the traffic lights. At that point the scooter starts to slow at a constant rate of 1 m s^{-2} until it stops.
- 4 The sketch shows a displacement–time graph of the position of a train passing a station. The displacement is measured from the entrance of the station to the front of the train. Find the equation of the displacement–time graph and hence the time at which the front of the train reaches the entrance of the station.



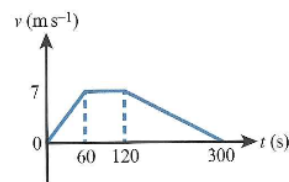
- 5 The sketch shows a displacement–time graph of a car slowing down with constant acceleration before coming to rest at a set of traffic lights.
- The equation of the displacement–time graph can be written in the form $s = p(t - q)^2 + r$. Using the two points marked and the fact that the car is stationary at $t = 10$, find p , q and r .
 - By comparison with the equation $s = s_0 + ut + \frac{1}{2}at^2$, find the initial speed and acceleration of the car.



- 6 Two cars drive along the same highway. One car starts at junction 1, travelling north at a constant speed of 30 m s^{-1} . The second car starts at junction 2, which is 3 km north of junction 1, travelling south at a constant speed of 20 m s^{-1} .
- Sketch the two displacement–time graphs on the same set of axes.
 - Find the equations of the two displacement–time graphs.
 - Solve the equations to find the time at which the cars pass each other and, hence, find the distance from junction 1 at which they pass.
- 7 Two trains travel on parallel tracks that are 5 km long. One starts at the southern end, travelling north at a constant speed of 25 m s^{-1} . The second train starts at the northern end 40 s later, travelling south at a constant speed of 15 m s^{-1} .
- Sketch the two displacement–time graphs on the same set of axes.
 - Find the time for which the first train has been moving and the distance the first train has travelled when the trains pass each other.

e. Velocity-time graphs

- 6 The graph shows the journey of a cyclist travelling in a straight line from home to school. Find the distance between her home and the school.



- 7 A racing car is being tested along a straight 1 km course. It starts from rest, accelerating at a constant rate of 10 m s^{-2} for 5 s. It then travels at a constant speed until a time t s after it started moving. Show that the distance covered by time t is given by $s = 125 + 50(t - 5)$. Hence, find how long it takes to complete the course.
- 8 A rowing boat accelerates from rest at a constant rate of 0.4 m s^{-2} for 5 s. It continues at constant velocity for some time until decelerating to rest at a constant rate of 0.8 m s^{-2} . In total, the boat covers a distance of 30 m. Find how long was spent at constant speed.
- 9 A cyclist accelerates at a constant rate for 10 s, starting from rest and reaching a speed of $v \text{ m s}^{-1}$. She then remains at that speed for a further 20 s. At the end of this she has travelled 300 m in total. Find the value of v .
- 10 A boat accelerates from rest at a rate of 0.2 m s^{-2} to a speed $v \text{ m s}^{-1}$. It then remains at that speed for a further 30 s. At the end of this it has travelled 400 m in total.
- Find the value of v .
 - What assumptions have been made to answer the question?
- 11 A crane lifts a block from ground level at a constant speed of $v \text{ m s}^{-1}$. After 5 s the block slips from its shackles and decelerates at 10 m s^{-2} . It reaches a maximum height of 6 m. Find the value of v .
- 12 A car is at rest when it accelerates at 5 m s^{-2} for 4 s. It then continues at a constant velocity. At the instant the car starts moving, a truck passes it, moving at a constant speed of 22 m s^{-1} . After 10 s the truck starts slowing at 1 m s^{-2} until coming to rest.
- Show that the velocities are equal after 12 s and, hence, find the maximum distance between the car and the truck.
 - Show that the distance covered at a time t s from the start by the car and the truck, for $t > 10$, are given by $40 + 20(t - 4)$ and $220 + 22(t - 10) - \frac{1}{2}(t - 10)^2$, respectively. Hence, find the time at which the car passes the truck.
- 13 Two cyclists are having a race along a straight road. Bradley starts 50 m ahead of Chris. Bradley starts from rest, accelerates to 15 m s^{-1} in 10 s and remains at this speed for 40 s before decelerating at 0.5 m s^{-2} . Chris starts 5 s later than Bradley. He starts from rest, accelerates to 16 m s^{-1} in 8 s and maintains this speed.
- Show that Bradley is still ahead when he starts to slow down, and find how far ahead he is.
 - Find the amount of time Bradley has been cycling when he is overtaken by Chris.
- 14 A driver travelling at 26 m s^{-1} sees a red traffic light ahead and starts to slow at 3 m s^{-2} by removing her foot from the accelerator pedal. A little later she brakes at 5 m s^{-2} and comes to rest at the lights after 6 s.
- Sketch the velocity–time graph of the motion.
 - Find the equations of the two sections of the graph.
 - Hence, find the time when the driver needs to start braking.

f. Graphs with discontinuities

- 1 An ice hockey puck slides along a rink at a constant speed of 10 m s^{-1} . It strikes the boards at the edge of the rink 20 m away and slides back along the rink at 8 m s^{-1} until going into the goal 40 m from the board. Sketch a velocity–time graph and a displacement–time graph for the motion, measuring displacement from the starting point in the original direction of motion.
- 2 A bowling ball rolls down an alley with initial speed 8 m s^{-1} and decelerates at a constant rate of 0.8 m s^{-2} . After 2.5 s it strikes a pin and instantly slows down to 2 m s^{-1} . It continues to decelerate at the same constant rate until coming to rest. Sketch a velocity–time graph and a displacement–time graph for the motion.
- 3 In a game of blind cricket, a ball is rolled towards a player with a bat 20 m away, who tries to hit the ball. On one occasion, the ball is rolled towards the batsman at a constant speed of 4 m s^{-1} . The batsman hits the ball back directly where it came from with initial speed 6 m s^{-1} and decelerating at a constant rate of 0.5 m s^{-2} . Sketch a velocity–time graph and a displacement–time graph for the motion, taking the original starting point as the origin and the original direction of motion as positive.
- 4 A ball is dropped from rest 20 m above the ground. It accelerates towards the ground at a constant rate of 10 m s^{-2} . It bounces on the ground and leaves with a speed that is half the speed it struck the ground originally. The ball is then caught when it reaches the highest point of its bounce. Sketch a velocity–time graph and a displacement–time graph for the motion, measuring displacement above the ground.
- 5 A ball is thrown towards a wall, which is 5 m away, at 2.25 m s^{-1} . It slows down at a constant rate of 0.2 m s^{-2} until it strikes the wall. It bounces back at 80% of the speed it struck the wall originally. It again slows down at a constant rate of 0.2 m s^{-2} until coming to rest. Sketch a velocity–time graph and a displacement–time graph for the motion, measuring displacement from the wall, taking the direction away from the wall as positive.
- 6 A billiard ball is on the centre spot of a 6 m long table and is struck towards one of the cushions with initial speed 3.1 m s^{-1} . It slows on the table at 0.2 m s^{-2} . When it bounces off the cushion its speed reduces to 70% of the speed with which it struck the cushion. The ball is left until it comes to rest.
 - a Sketch the velocity–time and displacement–time graphs for the ball, taking the centre of the table as the origin for displacement and the original direction of motion as positive.
 - b What assumptions have been made in your answer?
- 7 A ball is released from rest 20 m above the ground and accelerates under gravity at 10 m s^{-2} . When it bounces its speed halves. If bounce n occurs at time t_n the speed after the bounce is v_n . Show that $v_n = 15 - 2.5t_n$ and deduce that, despite infinitely many bounces, the ball stops bouncing after 6 s.

2. Force and motion

a. Newton's first law

- 7 A ship of mass 20 tonnes is moving at 10 m s^{-1} when its engines stop and it decelerates in the water. It takes 500 m to come to rest. Find the resistance force, which is assumed to be constant, of the water on the ship.
 - 8 A winch provides a constant force of 80 N and causes a block to accelerate on horizontal land from 2 m s^{-1} to 10 m s^{-1} in 6 s. Find the mass of the block.
 - 9 A Formula 1 car and its driver have total mass of 800 kg. The driver is travelling at 100 m s^{-1} along a horizontal straight when, 100 m before a bend, he starts to brake, slowing to 40 m s^{-1} . Assuming a constant braking force, find the force of the brakes on the car.
 - 10 A strongman drags a stone ball from rest along a horizontal surface. He moves it 10 m in 4 s by exerting a constant force of 100 N. Find the mass of the stone ball.
 - 11 A car and driver, of total mass 1350 kg, are moving at 30 m s^{-1} on a horizontal road when the driver sees roadworks 400 m ahead. She brakes, decelerating with a constant force of 600 N until arriving at the roadworks. Find the time elapsed before arriving at the roadworks.
 - 12 A train travelling at 30 m s^{-1} on a horizontal track starts decelerating 360 m before coming to rest at a platform. The brakes provide a constant resistance of 100 kN. Find the mass of the train.
 - 13 A block is being dragged along a horizontal surface by a constant horizontal force of size 45 N. It covers 8 m in the first 2 s and 8.5 m in the next 1 s. Find the mass of the block.
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b. Combinations of forces

- 4 A cyclist and her bike have a combined mass of 80 kg. She exerts a driving force of 200 N and experiences air resistance of 150 N. Find the acceleration of the cyclist.
- 5 A car of mass 1500 kg experiences air resistance of 450 N. It accelerates at 3 m s^{-2} on horizontal ground. Find the driving force exerted by the engine.
- 6 A boat of mass 2 tonnes has a driving force of 1000 N and accelerates at 0.2 m s^{-2} . Find the resistance that the water provides.
- 7 A water-skier of mass 75 kg is towed by a horizontal rope with constant tension of 150 N. There is constant resistance from the water of 120 N. Find the time taken to reach a speed of 10 m s^{-1} from rest.
- 8 A learner driver is being tested on the emergency stop. The car has mass 1250 kg and is moving at 21 m s^{-1} . When the driver presses the brakes there is a braking force of 10 000 N in addition to the air resistance of 500 N. Find the distance covered in coming to rest.

- 9 A small aircraft is accelerating on a runway. Its engines provide a constant driving force of 20 kN. There is average air resistance of 1000 N. The aircraft starts from rest and leaves the runway, which is 900 m long, with speed 80 m s^{-1} .
- Find the mass of the aircraft.
 - What assumptions have been made to answer the question?
- 10 A drag racing car races on a track of length 400 m. The car has mass 600 kg and accelerates from rest with a constant driving force of 21 kN. There is air resistance of 1800 N. Find the speed of the car at the finish line of the race.
- 11 A wooden block of mass 6 kg is being dragged along a horizontal surface by a force of 10 N. It accelerates from 1 m s^{-1} to 3 m s^{-1} in 4 s. Find the size of the friction force acting on the wooden block.
- 12 A motorcyclist is travelling at 20 m s^{-1} on level road when she approaches roadworks and slows down to 10 m s^{-1} over a distance of 250 m. The combined mass of the motorcyclist and the motorcycle is 360 kg. There is air resistance of 80 N. Find the braking force of the motorcycle.

c. Weight and motion due to gravity

- 7 A coin of mass 0.05 kg is dropped from the top of the Eiffel Tower, which is 300 m tall. It experiences air resistance of 0.01 N. Find the speed with which the coin hits the ground.
- 8 A winch lifts a bag of sand of mass 12 kg from the ground with a constant force of 240 N until it reaches a speed of 10 m s^{-1} . Then the winch provides a force to keep the bag moving at constant speed. Find the time taken to reach a height of 40 m.
- 9 A firework of mass 0.4 kg is fired vertically upwards with initial speed 40 m s^{-1} . The firework itself provides a force of 2 N upwards. The firework explodes after 6 s. Find the height at which it explodes.
- 10 A flare of mass 0.5 kg is fired vertically upwards with speed 30 m s^{-1} . The flare itself provides a force of 0.8 N upwards, even when the flare is falling, to keep the flare high for as long as possible. The flare is visible over the horizon when it reaches a height of 25 m.
- Find how long the flare is visible for.
 - What assumptions have you made in your answer?
- 11 A feather of mass 10 g falls from rest from a height of 3 m and takes 2 s to hit the ground. Find the air resistance on the feather.
- 12 A ball of mass 0.3 kg is thrown upwards with speed 10 m s^{-1} . It experiences air resistance of 0.15 N. The ball lands on the ground 1.2 m below. Find the speed with which it hits the ground.
- 13 A bouncy ball is dropped from a height of 5 m. When it bounces its speed immediately after impact is 80% of the speed immediately before impact.
- Find the maximum height of the ball after bouncing.
 - Show that the height is independent of the value used for g .
- 14 A parachutist of mass 70 kg falls out of an aeroplane from a height of 2000 m and falls under gravity until 600 m from the ground when he opens his parachute. The parachute provides a resistance of 2330 N. Find the speed at which the parachutist is travelling when he reaches the ground.



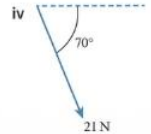
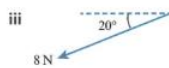
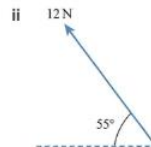
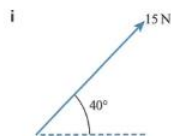
d. Normal contact force

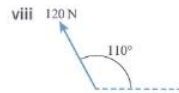
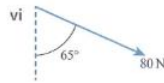
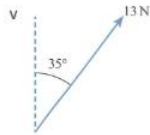
- 4 An elevator is carrying a child of mass 40 kg downwards. It is travelling at 8 m s^{-1} and starts to slow down at constant acceleration when it is 6.25 m from where it stops. Find the size of the normal contact force on the child.
- 5 A forklift truck carries a wooden pallet. On the pallet is a box of tiles with mass 35 kg. The truck lifts the pallet and tiles with an initial acceleration of 2 m s^{-2} . Find the normal contact force on the tiles.
- 6 A weightlifter is trying to lift a bar with mass 200 kg from the floor. He lifts with a force of 1800 N but cannot lift it off the floor. Find the size of the normal contact force from the floor on the bar while the weightlifter is trying to lift the bar.
- 7 A plate of mass 0.7 kg is being held on a horizontal tray. The tray is lifted from rest on the floor and accelerates at a constant rate until it reaches a height of 1.25 m after 5 s. Find the normal contact force on the plate.
- 8 A man of mass 75 kg is standing in the basket of a hot-air balloon. The balloon is rising at 5 m s^{-1} and 4 s later it is descending at 3 m s^{-1} . Assuming constant acceleration, find the normal contact force on the man.
- M** 9 A girl of mass 45 kg is sitting in a helicopter. The helicopter rises vertically with constant acceleration from rest to a speed of 40 m s^{-1} when it reaches a height of 200 m.
- Find the normal contact force on the girl.
 - What assumptions have been made to answer the question?



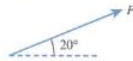
e. Resolving forces in horizontal and vertical directions

- 1 Find the components of the forces in the diagrams:
- horizontally, specifying whether it is left or right
 - vertically, specifying whether it is upwards or downwards.

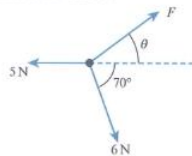




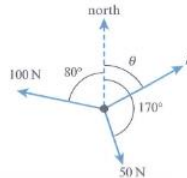
- 2 a A force, F , has a horizontal component, F_x , of 10 N and acts at 20° above the rightwards horizontal, as shown in the diagram. Find F and the vertical component, F_y .



- b A force, F , has a vertical component, F_y , of 8 N and acts at 25° to the right of the upwards vertical. Find F and the horizontal component, F_x .
- c A force, F , has a vertical component, F_y , of 8 N and a horizontal component, F_x , of 10 N. Find F and the angle, θ , that the force makes with the rightwards horizontal.
- d A force of 25 N has a horizontal component, F_x , of 17 N and acts above the horizontal. Find the vertical component, F_y , and the angle, θ , above the rightwards horizontal at which the force acts.
- e A force of 3.8 N has a vertical component, F_y , of 3 N and acts to the left of the vertical. Find the horizontal component, F_x , and the angle, θ , above the leftwards horizontal at which the force acts.
- 3 A particle in equilibrium has three forces of magnitudes 5 N, 6 N and F N acting on it in the horizontal plane in the directions shown. Find the values of F and θ .

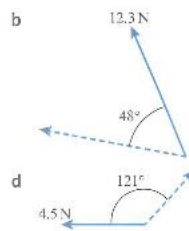
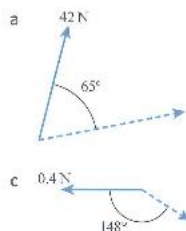


- 4 A lightshade of mass 2 kg is hung from the ceiling by two strings. One is fixed with tension 8 N at 20° to the vertical. The other is fixed with tension T N at an angle θ to the vertical.
- a By modelling the lightshade as a particle, draw a force diagram for this situation.
- b Resolve horizontally to find a value for $T \sin \theta$ and resolve vertically to find a value for $T \cos \theta$.
- c Hence, find the values of T and θ .
- 5 A ship is being blown by a breeze with a force of 100 N on a bearing of 280° , as shown in the diagram. It is pulled by a rope attached to the shore with force 50 N on a bearing of 170° . A tugboat holds it in place. Find the size and bearing of the force F applied by the tugboat.

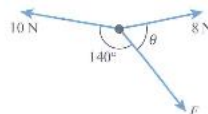


f. Resolving forces at equilibrium

- 2 Find the components of the following forces perpendicular to the direction of the dashed arrow. Make it clear whether the component is in the perpendicular direction clockwise or anticlockwise from the direction given.



- 3 A particle has three forces acting on it, as shown in the diagram. By resolving perpendicular to and parallel to F , find F and θ .

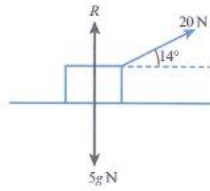


- 4 A boat is held in equilibrium by two tugboats. One pulls with a force of 100 N on a bearing of 190° . One pulls on a bearing of 340° with tension T . The wind blows with a force on the boat of F on a bearing of 50° . By resolving perpendicular to T , find T . Find also F .

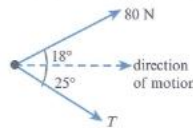
- 5 A book of mass 3 kg is prevented from sliding down a slope at 15° to the horizontal by friction acting up the slope and parallel to it. Find the force of friction and the normal contact force.
- 6 A wooden block of mass 4 kg is held at rest on a slope at angle θ to the horizontal by a force of 12 N acting up the slope and parallel to it. Find the slope's angle and the normal contact force.
- 7 A particle of mass 2 kg is held in equilibrium on a slope at 13° to the horizontal by a force F acting at 10° to the slope above it. Find F and the normal contact force.
- 8 A box of mass 12 kg is held in equilibrium on a slope at 18° to the horizontal by a force of size 50 N acting at an angle θ above the slope. Find θ and the normal contact force.
- 9 A boy is dragging a bag of mass 8 kg up a slope at an angle of 17° to the horizontal and exerts a force of 50 N parallel to the slope to do this. Air resistance, F , parallel to the slope prevents the boy from increasing his speed, so he maintains a constant speed. Find the magnitude of the air resistance and the normal contact force.
- 10 A girl is dragging a sled of mass 20 kg up a slope at angle 14° to the horizontal. She pulls at an angle of θ above the slope with a force of 70 N. She maintains a constant speed despite friction of 10 N parallel to the slope. Find θ and the normal contact force.

g. Resolving forces not in equilibrium

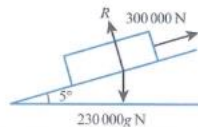
- 1 A wooden block of mass 5 kg is on a horizontal surface. It is dragged by a force of 20 N acting at 14° above the horizontal, as shown in the following diagram. Find the acceleration of the block and the normal contact force.



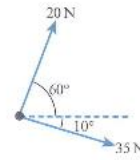
- 2 A book of mass 2 kg is dragged along a horizontal surface by a rope at 6° above the horizontal. It accelerates at 0.3 m s^{-2} .
- Draw the force diagram for this situation.
 - Find the tension in the rope and the normal contact force.
- 3 A box of mass 10 kg is pulled along a horizontal surface by a rope with tension 20 N at an angle θ above the horizontal. The box accelerates at 1.2 m s^{-2} . Find θ and the normal contact force.
- 4 A car of mass 1000 kg is being towed by two people holding ropes. One pulls with a tension of 80 N at an angle of 18° to the direction of motion. The other pulls at an angle of 25° to the direction of motion, as shown in the diagram. Find the tension, T , in the second rope and the acceleration of the car.



- 5 A box of mass 20 kg is dragged by a force of 40 N at an angle of 15° to the direction of motion, and a force of 30 N at an angle of θ to the direction of motion. Find the value of θ and the acceleration of the box.
- 6 A truck of mass 15000 kg is being towed by two ropes. One pulls with a tension of 3000 N at an angle of 20° to the direction of motion. The other pulls with a tension, T , at an angle of 10° to the direction of motion. There is resistance of 500 N against the motion, in the same line as the motion.
- Draw the force diagram for this situation.
 - Find T and the acceleration of the truck.
- 7 A ship of mass 10000 kg is being towed due north by two tugboats with acceleration 0.1 m s^{-2} . One pulls with a tension of 2000 N on a bearing of 330° . The other pulls with a tension, T , on a bearing of θ . There is resistance against the motion of 1000 N. Find T and θ .
- 8 A train of mass 230 tonnes provides a driving force of 300000 N to accelerate up a slope at an angle of 5° to the horizontal. The force diagram is shown. Find the acceleration of the train.



- 4 A particle of mass 25 kg has two forces acting on it, one of 20 N and one of 35 N, in the directions shown. Find the magnitude and direction of the resulting acceleration.

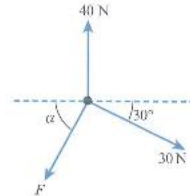


- 5 Three coplanar forces act on a particle, as shown in the following diagram. X has components 0 N in the x -direction and 20 N in the y -direction. Y has components 25 N in the x -direction and -10 N in the y -direction. Z has components -10 N in the x -direction and -15 N in the y -direction. Find the magnitude and direction of the resultant of the three forces.

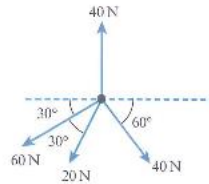


- 6 Three coplanar forces act on a particle, as shown in the diagram.

- Force F has components of -30 N in the x -direction and -40 N in the y -direction. Find the value of α .
- Find the magnitude and direction of the resultant of the three forces.

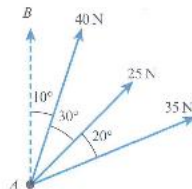


- 7 Three coplanar forces act on a particle, as shown in the following diagram. Show that the x component and y component of the resultant are equal. Hence, determine the direction of the resultant force.



- 8 Three people drag a bag of sand of mass 150 kg, labelled A in the diagram. They pull in the horizontal plane with forces 40 N, 25 N and 35 N in the directions shown, compared to the direction AB .

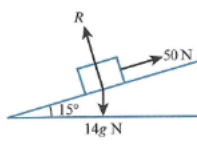
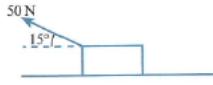
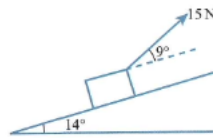
- Find the magnitude and direction of acceleration of the resultant motion.
- What assumptions have been made to answer the question?



- 9 A boat of mass 1000 kg is pulled by three tugboats. One pulls due north with force 500 N, one pulls due east with force 350 N and one pulls on a bearing of 040° with a force of 250 N. Find the bearing and acceleration of the resultant motion.

3. Friction

a. Friction as contact force

- 2 A box of mass 14 kg is at rest on a slope that is at 15° to the horizontal.
- When there is no external force acting on the box, find the size and direction of the force of friction.
 - When it is pulled up the slope by a force of 50 N parallel to the line of greatest slope, as shown in the diagram, find the size and direction of the force of friction. (Note that friction is not marked. You will have to decide which direction you think friction is acting and come to a conclusion based on whether the answer you get is positive or negative.)
 - When the box is dragged down the slope by a force of 20 N at 10° above the line of greatest slope, find the size and direction of the force of friction.
 - When it is pulled up the slope by a force of 15 N at 35° above the horizontal, find the size and direction of the force of friction.
- 
- 3 A box of mass 20 kg is at rest on rough horizontal ground. Find the magnitude of the total contact force in each of these cases.
- The box is pulled horizontally to the right by a force of 40 N .
 - The box is pushed to the left by a force of 50 N at 15° above the horizontal, as shown in the diagram.
 - The box is pushed to the left by a force of 50 N at 15° below the horizontal.
- 
- 4 A book of mass 4 kg is at rest on a rough slope at angle 14° to the horizontal. Find the magnitude of the total contact force in each of these cases.
- No other force acts on the book.
 - The book is pulled down the slope by a force of 5 N parallel to the line of greatest slope.
 - The book is pulled up the slope by a force of 15 N at 9° above the line of greatest slope, as shown in the diagram.
- 
- 5 A tin of mass 0.5 kg is on a rough horizontal table with coefficient of friction 0.3 . Find the largest horizontal force that can be exerted on the tin before the tin starts to move.
- 6 A block of wood of mass 3 kg is on a rough slope, which is at an angle of 25° to the horizontal. The coefficient of friction between the block and the slope is 0.4 . It is held in place by a force, P , going up the line of greatest slope.
- Find the smallest possible size of P to prevent the block sliding down the slope.
 - Given that the block remains in equilibrium, find the largest possible size of P .
- 7 A chair of mass 6 kg is at rest on a rough horizontal floor with coefficient of friction 0.35 . It is pulled horizontally by a force of 25 N . A boy pushes down on the chair so that the chair is on the point of slipping but remains at rest. Find the force that the boy exerts on the chair.
- 8 Two men are trying to drag a bin of mass 100 kg up a rough slope at an angle 20° to the horizontal. The coefficient of friction is 0.25 . One man pulls up the slope with a force of 400 N . The other tries to lift the bin perpendicularly to the slope, providing a force such that the bin is on the point of slipping up the slope. Find the force exerted by the second man.
- 9 A sledge of mass 200 kg is being pulled by a woman along rough horizontal ground. She exerts a force of 500 N at 18° above the horizontal and the sledge is on the point of slipping. Find the coefficient of friction.

b. Limit of friction

- 4 A gardener is pulling a wheelbarrow of mass 8 kg from rest along rough horizontal ground. The coefficient of friction between the wheelbarrow and the ground is 0.6. The gardener provides a force of 50 N at an angle of 30° above the horizontal, as shown in the diagram.



- a Find the acceleration of the wheelbarrow.
 - b What happens when the wheelbarrow has 20 kg of soil in it and the gardener exerts the same force at the same angle?
- 5 A ski-plane has skis to land and take-off on snow. It has a mass of 3000 kg and has a propeller providing a force of 20 000 N horizontally. It accelerates from rest on horizontal ground at 2.2 m s^{-2} . Find the coefficient of friction between the ground and the ski-plane.
 - 6 A bin of mass 16 kg is held on a downhill slope at an angle of 20° . When the bin is released, it slides down the slope with acceleration 1.2 m s^{-2} . Find the coefficient of friction between the bin and the ground.
 - 7 A ring of mass 2 kg is on a fixed rough horizontal wire with coefficient of friction 0.4. It is pulled by a rope with tension 15 N at an angle of 5° above the horizontal. Find the acceleration of the ring.
 - 8 A ring of mass 3 kg is on a fixed rough horizontal wire. It is pulled by a rope with tension 20 N at an angle of 10° above the horizontal and accelerates at 2 m s^{-2} . Find the coefficient of friction between the ring and the wire.
 - 9 A ski-plane of mass 5000 kg accelerates from rest along a rough horizontal runway of length 600 m. It needs to reach a speed of 25 m s^{-1} by the end of the runway to take off. The propeller provides a horizontal force of 16 000 N. Find the maximum coefficient of friction to allow the ski plane to take off.
 - 10 A downhill skier of mass 80 kg is accelerating down a rough slope of length 400 m at 22° to the horizontal. There is air resistance of 50 N and the coefficient of friction between the snow and the skis is 0.3. The skier is moving at 20 m s^{-1} at the top of the slope. Find the speed of the skier at the bottom of the slope.
 - 11 A bag of sand of mass 200 kg is being winched up a slope of length 10 m, which is at an angle of 6° to the horizontal. The slope is rough and the coefficient of friction is 0.4. The winch provides a force of 1000 N parallel to the slope. At the bottom of the slope the bag is moving at 2 m s^{-1} . Find the distance it has moved when its speed has reduced to 1.5 m s^{-1} .
 - 12 A man wants to drag a block of wood of mass 50 kg along horizontal rough ground, where the coefficient of friction is 0.45. If he pushes it he can generate a force of 250 N horizontally. Alternatively, he can pull via a string with a force of only 230 N at an angle of 25° above the horizontal. Which would give the larger acceleration?
 - 13 Two men are pushing a palette of bricks of mass 120 kg along rough horizontal ground. The first man pushes horizontally with a force of 150 N. The second man pulls via a rope at an angle of 20° above the horizontal with a force of 140 N. They maintain a constant velocity.
 - a Find the coefficient of friction between the palette and the ground.
 - b The second man no longer pulls the rope. By first finding the new normal contact force, find the deceleration of the palette.
 - 14 A snooker ball of mass 0.4 kg is struck towards a cushion from 0.8 m away with speed 3 m s^{-1} . The surface of the snooker table has a coefficient of friction of 0.3. When the ball bounces from the cushion its speed is reduced by 20%. Find how far from the cushion it stops.

c. Changes of direction with relation to friction

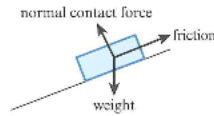
- 1 At the end of a downhill run, a skier of mass 80 kg slides up a rough slope at an angle of 10° to the horizontal, to slow down. He arrives at the upward slope with an initial speed of 12 m s^{-1} . The coefficient of friction between the skier and the slope is 0.4. Find how far up the slope he comes to rest, and show that he remains at rest there without falling back down the slope.
 - 2 A ball of mass 3 kg rolls with initial speed 8 m s^{-1} up a rough slope at an angle of 15° to the horizontal. The coefficient of friction between the ball and the slope is 0.6.
 - a By modelling the ball as a particle, find how long it takes for the ball to come to rest and show that the ball remains at rest there.
 - b Why is this model different from reality?
 - 3 A book of mass 3 kg is at rest on a rough slope at an angle of 15° to the horizontal. It takes a force of 20 N parallel to the slope to break equilibrium and drag it up the slope.
 - a Find the coefficient of friction between the slope and the book.
 - b Find the acceleration of the book down the slope if the 20 N force is applied down the slope.
 - 4 A box of mass 12 kg is at rest on a rough slope at an angle of 18° to the horizontal. The coefficient of friction between the slope and the box is 0.4.
 - a Find the force it takes parallel to the upwards slope to break equilibrium and drag the box up the slope.
 - b If the force were applied down the slope and parallel to it, find the acceleration.
- PS**
- 5 A car of mass 1250 kg is at rest on a rough slope at an angle of 35° to the horizontal. It takes a force of 13 000 N to move it up the slope. Show that without any force the car would slide down the slope, and find the minimum force to prevent it moving down.
 - 6 A bin of mass 10 kg is at rest on a rough slope at an angle of 32° to the horizontal. It is held on the point of moving up the slope by a force of 90 N parallel to the slope. Show that when the force is removed the bin would slide down the slope, and find its acceleration.
 - 7 A trolley of mass 5 kg is rolling up a rough slope, which is at an angle of 25° to the horizontal. The coefficient of friction between the trolley and the slope is 0.4. It passes a point *A* with speed 12 m s^{-1} . Find its speed when it passes *A* on its way back down the slope.
 - 8 A ball of mass 1.5 kg is sliding up a slope, which is at 30° to the horizontal. The coefficient of friction between the ball and the slope is 0.45. It passes a point *A* at 10 m s^{-1} . By modelling the ball as a particle, find the time taken to return to *A*.
- M**
- 9 A pinball game involves hitting a ball up a slope whenever it reaches the bottom of the slope. The pinball has mass 0.2 kg and rolls down a rough slope of length 1.2 m at angle 12° to the horizontal and with coefficient of friction 0.1. The ball starts at the top of the slope at rest. When it reaches the bottom of the slope it is hit back up and its speed is increased by 50%.
 - a Find the maximum height up the slope the pinball reaches after it has been hit back up the slope.
 - b What assumptions have been made to answer the question?
 - 10 A wooden block of mass 3.5 kg is sliding up a rough slope and passes a point *A* with speed 20 m s^{-1} . The slope is at 29° to the horizontal. The block comes to rest 25 m up the slope. Find its speed as it passes point *A* on the way down.
 - 11 A boy drags a sledge of mass 4 kg from rest down a rough slope at an angle of 18° to the horizontal. He pulls it with a force of 8 N for 3 s by a rope that is angled at 10° above the parallel down the slope. After 3 s the rope becomes detached from the sledge. The coefficient of friction between the slope and the sledge is 0.4. Find the total distance the sledge has moved down the slope from when the boy started dragging it until it comes to rest.
 - 12 A particle slides up a slope at angle 34° to the horizontal with coefficient of friction 0.4. It passes a point *P* on the way up the slope with speed 3 m s^{-1} and passes it on the way down the slope with speed 2 m s^{-1} . Find the coefficient of friction between the particle and the slope.

4. Connected particles

a. Newton's third law

A box resting on the floor pushes down on the floor with a vertical contact force and the floor pushes up on the box with an equal and opposite contact force. If no other forces act, these contact forces are each equal to the weight of the box.

A box resting on a slope pushes into the slope with a contact force and the slope pushes back with an equal and opposite contact force. When you draw a force diagram, you usually show only the forces acting *on* the box, so here you show the normal contact force from the slope on the box.



b. Objects connected by rods

- 6 A bucket hangs from a vertical rod. Another rod is attached to the bottom of the bucket and a second bucket hangs on the end of this rod. Each bucket is partially filled with water and they hang in equilibrium.
- a Work out the tension in each rod when:
- each bucket of water has mass 12 kg
 - the first bucket of water has mass 8 kg and the second has mass 16 kg.
- b What assumptions have been made?
- 7 A 15 kg mass hangs in equilibrium on a heavy chain. The chain is modelled as ten 1 kg masses joined by short rods of negligible mass. Including the connection at each end of the chain, this makes 11 short rods. Work out the tension in each of the 11 short rods.
- 8 A horizontal bar of mass 1 kg hangs from a pair of parallel vertical rods, of negligible mass, attached to either end of the bar. A third vertical rod is connected to the middle of the bar and a 4 kg mass hangs from this, below the rod. Work out the tension in each of the rods.
- 9 A train consists of an engine and five carriages. The engine has mass 100 000 kg and each carriage has mass 20 000 kg. The engine produces a driving force of 350 000 N. The resistance force on the engine is 10 000 N and the resistance on each carriage is 2000 N. The train moves in a straight line on a horizontal track. Find the tension in the coupling between the third carriage and the fourth carriage.
- 10 A car pulls a caravan up a hill. The slope of the hill makes an angle θ with the horizontal, where $\sin \theta = \frac{1}{20}$. The car has mass 1900 kg and the caravan has mass 600 kg. The driving force from the engine of the car is 1200 N. The resistance on the car is 20 N and that on the caravan is 80 N. Find the force in the tow-bar and state whether it is a tension force or a thrust force.
- 11 A car tows a caravan down a hill. The slope of the hill makes an angle θ with the horizontal, where $\sin \theta = \frac{1}{20}$. The car has mass 1900 kg and the caravan has mass 600 kg. The car is braking, so the driving force from the engine of the car is negative. This braking force is 250 N (a driving force of -250 N). The resistance on the car is 20 N and that on the caravan is 80 N. Find the force in the tow-bar and state whether it is a tension force or a thrust force.
- 12 A car tows a caravan down a hill. The slope of the hill makes an angle θ with the horizontal, where $\sin \theta = 0.05$. The force from the car's engine is a braking force (a negative driving force). The car has mass 1800 kg and the caravan has mass 600 kg. The resistance on the car is 20 N and that on the caravan is 80 N. The force in the tow-bar is a thrust of 50 N. Show that the force from the car's engine is -420 N.

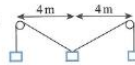
c. Objects connected by strings

- 3 A light inextensible string is fixed to a point on a ceiling. A box of mass 2 kg hangs from the string. Two light inextensible strings are attached to the box and hang vertically below the box. A particle of mass 0.4 kg hangs from the lower end of one string and a particle of mass 0.6 kg hangs from the lower end of the other string. Work out the tension in each string.
- 4 A block of mass 5 kg hangs from one end of a light inextensible string of length 2 m. The string passes over a small smooth pulley at the edge of a smooth horizontal table of height 70 cm. The other end of the string is connected to a block of mass 3 kg held at rest on the table. The portion of the string between the pulley and the 3 kg mass is horizontal and of length 1.5 m. The system is released from rest.
- How long does it take for the 5 kg mass to reach the ground?
 - What is the speed of the 3 kg mass when the 5 kg mass hits the ground?
- The 3 kg mass continues to slide towards the pulley.
- How long does it take, from when the system is released, for the 3 kg mass to reach the pulley?
- 5 A mass X , of 1 kg, hangs from one end of a light inextensible string. The string passes over a small, smooth, fixed pulley and a mass Y , of 0.6 kg, hangs from the other end of the string. Initially X is 0.4 m below the pulley and Y is 1.8 m below the pulley. The pulley is 3 m above the ground. The system is released from rest.
- Find how long it takes from when the system is released until Y hits the pulley.
- When Y hits the pulley, the string breaks.
- Find how long it takes from when the system is released until X hits the ground.

- 6 A block of mass 4 kg is held on a rough slope that is inclined at 30° to the horizontal. The coefficient of friction between the slope and the block is 0.2. A light inextensible string is attached to the block and runs parallel to the slope to pass over a small smooth pulley fixed at the top of the slope. The other end of the string hangs vertically with a block of mass 1 kg attached at the other end. The system is released from rest.
- Work out the tension in the string.
- After 1.2 s the block of mass 4 kg reaches the bottom of the slope. The other block has not yet reached the pulley.
- Work out a lower bound for the length of the string, giving your answer to 2 significant figures.

- 7 A particle of mass 0.3 kg hangs from one end of a light inextensible string. The string passes over a smooth pulley and a particle of mass 0.5 kg hangs from the other end. A second string is tied to the particle of mass 0.5 kg, and a particle of mass 0.2 kg hangs from this string, so that the particle of mass 0.2 kg hangs vertically below the particle of mass 0.5 kg. The system is released from rest. Find the tension in each string.

- 8 Two smooth pulleys are 8 m apart at the same horizontal level. A light inextensible rope passes over the pulleys and a box of mass 5 kg hangs at each end of the rope. A third box of mass m kg is attached to the midpoint of the rope and hangs between the pulleys so that all three boxes are at the same horizontal level.



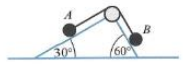
The total length of the rope is 16 m. Find the value of m .

- 9 Two smooth pulleys are fixed at the same horizontal level. A light inextensible rope passes over the pulleys and a box of mass m kg hangs at each end of the rope. A third box of mass km kg is attached to the midpoint of the rope and hangs between the pulleys so that all three boxes are at the same horizontal level. The portions of the string that are not vertical make an angle of 30° with the horizontal. The system hangs in equilibrium. Find the value of k .
- 10 A mass of 2 kg is held on a rough horizontal table. The coefficient of friction between the table and the mass is 0.05. The 2 kg mass is attached by a light inextensible string to a mass of 3 kg and by a second light inextensible string to a mass of 4 kg. The strings pass over smooth pulleys at the edges of the table. The 3 kg mass hangs on one side of the table and the 4 kg mass hangs on the other side of the table. The system is released from rest. Find:
- the acceleration of the masses
 - the tension in each string.

- PS 11 A wedge has two smooth sloping faces; one face makes an angle 30° with the horizontal and the other makes an angle 60° with the horizontal. A small smooth pulley is fixed at the apex of the wedge. A light inextensible string passes over the pulley and lies parallel to the faces of the wedge. At each end of the string there is a particle of mass 0.3 kg. The system is released from rest.

a Show that the tension in the string is $\frac{3}{4(1+\sqrt{3})}$ N.

b Work out the resultant horizontal force on each of the particles.



- PS 12 A box of mass 3 kg hangs from a light inextensible string, which passes over a smooth pulley fixed below a beam and then under a smooth cylinder of mass 4 kg that is free to move. The other end of the string is fixed to the beam.



The system is released from rest.

- Explain why the magnitude of the acceleration of the cylinder is half the magnitude of the acceleration of the box.
- Find the acceleration of the box, including its direction.

5. General motion in straight line

a. Velocity as derivative of displacement

- 4 A child on a fairground ride moves in a straight line. The position of the child, measured from the start, at time t s is given by $s = 0.5t^4 - t^2$ for $0 < t < 2$.
Find the speed of the child:
- when $t = 0$
 - when $t = 1$
 - when $t = 2$.
- 5 The position of a particle as it moves along a line is modelled as $s = 3 + 4t - t^2$, where s is the displacement, in metres, from a fixed point O and t is the time, in seconds, from the start.
- Show that the particle started 3 m from O .
 - Find how far the particle is from O when it is instantaneously at rest.
- 6 A tennis ball is projected vertically upwards. The vertical displacement of the ball, in metres, from the point of projection at time t seconds is given by $s = -5t^2 + 8t$.
- Find the time when the ball returns to its starting point.
 - Find the displacement when the ball is momentarily stationary.
- 7 A small stone is dropped into a lake. The stone descends vertically so that t s after entering the water it is s m below the surface of the water, where $s = 4t^2 - \sqrt{2}t^3$. The stone lands at the bottom of the lake with speed 13 m s^{-1} .
- Show that the stone takes 2 s to reach the bottom of the lake.
 - Work out the depth of the lake at the point where the stone lands.
- 8 In a drag race, two cars, A and B, line up side by side at the start point. When the starting flag is waved, both cars are driven as fast as possible in a straight line. The first car to cross the finish line is the winning car. At time t s from when the cars start to move, the distance travelled by car A is given by $s = 4t + t^2$. Car A takes 16 s to reach the finish line.
- Work out the distance from the start to the finish.
 - Find the speed of car A when it crosses the finish line.
- At time t s from when the cars start to move, the distance travelled by car B is given by $s = 1.2t^2$.
- Find the speed of car B when it crosses the finish line.
 - When the winning car crosses the finish line, how far behind it is the other car?

b. Acceleration as derivative of velocity

- 4 A car moves in a straight line. The velocity of the car, $v \text{ m s}^{-1}$, at time $t \text{ s}$ is given by $v = 5t + 0.5t^2$ for $0 < t < 2$. Find the acceleration of the car:
- when $t = 0$
 - when $t = 1$
 - when $t = 2$.

- 5 A boy runs in a straight line. The velocity of the boy, $v \text{ m s}^{-1}$, at time $t \text{ s}$ is given by:

$$v = \begin{cases} 2 + 3t & : 0 \leq t \leq 1 \\ 6 - t^2 & : 1 \leq t \leq 1.2 \\ 5 - \frac{k}{30}t & : 1.2 \leq t \leq T \end{cases} \quad \text{for } 0 \leq t \leq T$$

The boy has velocity 0 m s^{-1} at time T .

- Work out the value of k .
 - Show that T is just under 14 s .
 - Work out the average acceleration of the boy over the period $0-T$.
 - Show that there is no time at which the acceleration of the boy is the same as his average acceleration.
- 6 The motion of a cat, moving along a straight line, is modelled as $s = 6t - t^2$ for small values of t , where s is measured in metres and t in seconds.
- Find an expression for the velocity of the cat as a function of time.
 - Describe the motion of the cat.
 - When does the cat come to momentary rest?
 - Find the acceleration of the cat.
- 7 A particle moves forwards and backwards along a straight line. The velocity of the particle, $v \text{ m s}^{-1}$, at time $t \text{ s}$ is given by $v = -t^3 + 75t$ for $0 < t < 10$. Find the maximum speed of the particle.
- 8 A robot moves along a straight line for 3 s . The displacement of the robot, $s \text{ m}$, from its initial position is given by $s = At^3 + Bt^2 + Ct$ for constants A , B and C , where t is measured in seconds and $0 < t < 3$. The robot starts with velocity 2 m s^{-1} . It travels 6 m before coming to rest at time $t = 3$. Work out the values of A , B and C .
- 9 A particle moves along a straight line. The displacement of the particle, $s \text{ m}$, at time $t \text{ s}$ is given by $s = 4t^2 - 0.25t^3$ for $0 < t < 12$.
- Find the maximum speed of the particle.
 - Work out the difference between the time when the speed is greatest and the time when the speed is least.
- 10 A ball moves in a straight line. The velocity of the ball, $v \text{ m s}^{-1}$, at time $t \text{ s}$ is given by $v = 5 + 4t - t^2$ for $0 < t < 5$.
- Write down the initial velocity of the ball.
 - Work out the maximum velocity of the ball.
 - Find the average acceleration between the start and the end of the motion.

c. Displace as integral of velocity

- 5 A small stone is dropped into a well. It falls down the well, from rest, with no resistance for 2 s. It then hits the surface of the water and continues to fall vertically through the water until it reaches the bottom of the well. In the water the downwards velocity of the stone, $v \text{ m s}^{-1}$, is given by $v = 20 - t$, where t is the time, in seconds, measured from when the stone hits the surface of the water. The stone takes 2.5 s in total to reach the bottom of the well.
- Calculate the depth of the well.
 - If air resistance is taken into account, would you expect the depth of the well to be greater or smaller than your answer from part a?
- 6 A ball bearing is fired vertically upwards in a straight line through a tub of butter. The upward velocity of the ball bearing is given by $v = 13 - 10t - 3t^2 \text{ cm s}^{-1}$, where t is the time from when it was fired upwards.
- Find the time when the ball bearing comes momentarily to rest.
 - Find how far the ball bearing has travelled upwards at this time.
- The ball bearing then falls downwards through the hole it has made in the butter. The downward velocity of the ball bearing is given by $v = 10T \text{ cm s}^{-1}$, where T is the time from when it was momentarily at rest.
- Find the time that the ball bearing takes (from when it was momentarily at rest) to fall to its original position.
 - What assumptions have been made in the model used in part c, and how could the model be improved?
- 7 A particle moves on a straight line. The velocity of the particle, $v \text{ m s}^{-1}$, at time t s is given by $v = -t^3 + 9t \text{ m s}^{-1}$ for $0 < t < 5$.
- Find the displacement of the particle, from its original position, when $t = 5$.
 - Work out the distance that the particle travels from $t = 0$ to $t = 5$.
- 8 A car moves in a straight line with velocity $v \text{ m s}^{-1}$ at time t s, given by $v = 16 - 0.5t^{1.5} + t$ for $0 \leq t \leq 25$.
- Find the displacement of the car, from its original position, when $t = 25$.
 - Work out the distance that the car travels from $t = 0$ to $t = 25$. (Note: you will need an equation solver for this question. In the examination you will only be asked to solve the equations that can be done using algebraic methods.)
- 9 A ball rolls forwards and backwards in a long straight tube. The velocity, $v \text{ m s}^{-1}$, at time t s after measurement starts is given by:
- $$v = 6 + t^{0.5} \quad \text{for } 0 \leq t \leq 25$$
- $$v = A - t \quad \text{for } t \geq 25$$
- for some constant A .
- Show that $A = 36$.
 - Find the distance from the start when the ball changes direction.
- 10 A particle moves in a straight line, starting from rest. At time t s after the start, the velocity, $v \text{ m s}^{-1}$, of the particle is given by:
- $$v = 2t \quad \text{for } 0 \leq t \leq 4$$
- $$v = 9 - (t - 5)^2 \quad \text{for } 4 \leq t \leq 10$$
- Find the maximum speed of the particle.
 - Work out the distance that the particle moves in the time interval $0 \leq t \leq 10$.
- 11 Two cars travel towards one another on the two sides of a long straight road. They start 300 m apart and each car stops when its speed is 0 m s^{-1} . The time, in seconds, after the first car starts to move is t . The velocity, $v_1 \text{ m s}^{-1}$, of the first car is given by $v_1 = 7.5t - 0.5t^2$. The second car starts 2 s after the first, at $t = 2$. The velocity, $v_2 \text{ m s}^{-1}$, of the second car is given by $v_2 = (t - 2)^2 - 16$.
- Write down the initial speed of each car.
 - How far does the first car travel?
 - How far does the second car travel?
 - For what value of t are the cars alongside one another? (Note: you will need an equation solver for this question. You will not be allowed an equation solver in the examination.)

d. Velocity as integral of acceleration

- 5 A block slides down a sloped surface with a varying coefficient of friction. The acceleration, $a \text{ m s}^{-2}$, of the block (measured down the surface) is given by $a = 0.01(10t - 3)$, where t is the time in seconds. At $t = 0$ the block is at rest at the top of the surface. The block reaches the bottom of the surface with speed 2.24 m s^{-1} . How far does the block travel down the sloped surface?
- 6 A ball moves with acceleration given by $a = 0.01(4t + 3t^{0.5})$, where t is the time, in seconds. At $t = 1$ the ball is moving with velocity 0.5 m s^{-1} . Find the displacement of the ball between $t = 0$ and $t = 4$.
- 7 A robot moves in a straight line with acceleration $a \text{ m s}^{-2}$ at time $t \text{ s}$, given by $a = 40(t - 1)^3$. The minimum velocity of the robot in the subsequent motion is 0 m s^{-1} (the velocity is never negative).
- Show that the robot is stationary ($v = 0$) when $t = 1$.
 - Find the displacement of the robot, measured from the initial position, when the robot is stationary.
- 8 A particle starts at the origin and moves along the x -axis. The acceleration of the particle in the direction of the positive x -axis is $a = 6t - c$ for some constant c . The particle is initially stationary and it is stationary again when it is at the point with x -coordinate $= -4$. Find the value of c .
-
- 9 A goods train starts from rest at point A and moves along a straight track. The train moves with acceleration $a \text{ m s}^{-2}$ at time $t \text{ s}$, given by $a = 0.1t^2(6 - t)$ for $0 \leq t < 6$. It then moves at constant velocity for $6 \leq t < 156$ before decelerating uniformly to stop at point B at $t = 165$. Calculate the distance from A to B .
- 10 Two cars are travelling towards one another on the two sides of a long straight road. Each car stops when its speed is 0 m s^{-1} . The time, in seconds, after the first car starts to move is t . The acceleration, $a_1 \text{ m s}^{-2}$, of the first car is given by $a_1 = 5 - 2t$. The maximum velocity that the first car achieves is 26.01 m s^{-1} .
- Work out the initial velocity of the first car.
 - Find the time when the first car stops moving.
- The velocity, $v_2 \text{ m s}^{-1}$, of the second car is given by $v_2 = t^2 - 16$.
- How far does the second car travel?
- Initially the cars are 200 m apart.
- Show that the cars stop before they meet.
- 11 An ice hockey player hits the puck so that it moves across the ice in a horizontal straight line with acceleration $a \text{ m s}^{-2}$ at time $t \text{ s}$, where $a = -0.03t^2$. The initial speed of the puck, along the direction of motion, is 40 m s^{-1} .
- Find the distance that the puck travels in the first 2 seconds (between $t = 0$ and $t = 2$).
 - Find the speed of the puck after 2 seconds.
- When $t = 2$ the puck is stopped by an opposing player. This player then hits the puck back the way it came, giving it an initial speed of 30 m s^{-1} . The acceleration of the puck, in its direction of travel, is still given by $a = -0.03t^2$. The puck returns to its original starting point.
- Find, to 3 significant figures, how long it takes for the puck to return to its original starting point.
- 12 A girl bowls a ball along a straight and horizontal skittle alley. The forces acting on the ball are its weight, the normal contact force, friction and air resistance. The coefficient of friction between the ball and the surface of the skittle alley is 0.01 . The girl models the air resistance, in newtons, as $m(0.9 - 1.5t)$, where m is the mass of the ball, in kg , and t is the time, in s .
- Show that the velocity of the ball along the skittle alley, $v \text{ m s}^{-1}$, is given by $v = 0.75t^2 - t + C$ for some constant C .
- The initial velocity of the ball is 8 m s^{-1} . The skittle alley is 7.25 m long and the ball reaches the end of the skittle alley with velocity 2 m s^{-1} .
- Show that the ball takes just over 2.3 s to reach the end of the skittle alley.
- (Note: you will need an equation solver for this question. You will not be allowed an equation solver in the examination.)
- Why is the model for air resistance unreasonable?

6. Momentum

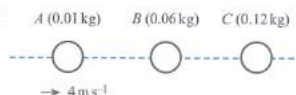
a. Momentum definition

- 6 A girl of mass 35 kg jumps from a rock onto the beach below. Her initial vertical speed is 0 m s^{-1} and she falls 2.45 m under gravity.
- Find the speed of the girl when she lands on the beach.
 - Find the downward momentum of the girl just before she lands on the beach.
- 7 A book of mass 2 kg falls from a window ledge and drops 1.8 m to the ground. It falls freely under gravity.
- Find the speed of the book just before it hits the ground.
 - Find the downward momentum of the book just before it hits the ground.
- 8 A ball of mass 0.2 kg falls 1.25 m vertically downwards to the ground, starting from rest. It hits the ground and rebounds. The downwards momentum of the ball changes by 1.6 N s in the bounce.
- What height does the ball reach after this bounce?
 - By considering the modelling assumptions, explain why the height might be less than this.
- 9 A ball bearing of mass 25 g is thrown vertically upwards, and is caught on the way back down. The ball bearing has an initial speed of 3 m s^{-1} upwards and is travelling at 2 m s^{-1} when it is caught. Find the change in its momentum.
- 10 A hockey ball of mass 0.2 kg is hit so that it has an initial speed of 8 m s^{-1} . The ball travels in a horizontal straight line with acceleration $a \text{ m s}^{-2}$ given by $a = -0.5 - kt$ where t is the time in seconds, measured from when the ball was hit. After 2 s the ball has travelled $\frac{41}{3} \text{ m}$. It is then intercepted by a player from the other team. This player hits the ball so that its direction of travel is reversed and its speed is now 5 m s^{-1} . Show that when the ball is hit by the second player its momentum changes in magnitude by 2 N s.

b. Collisions and conservation of momentum

- 3 A box of mass 25 kg slides down a slope until it has reached a speed of 5 m s^{-1} . It then travels at 5 m s^{-1} horizontally across a smooth floor until it bumps into a stationary crate. Immediately after the impact the box reverses its direction and travels at 0.5 m s^{-1} . The crate starts to travel at 2.75 m s^{-1} . Find the mass of the crate.
- 4 Two snooker balls are travelling towards one another in a straight line when they make a direct impact. Before the impact the first ball had speed 12 m s^{-1} and the second ball had speed 8 m s^{-1} . After the impact both balls have reversed their direction and each has speed 10 m s^{-1} . It is claimed that the balls are not both real snooker balls because they have different masses. Find the ratio of the masses of the balls.

- 5 Particles A , B and C , of masses 0.01 kg , 0.06 kg and 0.12 kg respectively, are at rest in a straight line on a smooth horizontal surface, with B between A and C . A is given an initial velocity of 4 m s^{-1} towards B . After this impact A rebounds with velocity 2 m s^{-1} and B goes on to hit C . After the second impact B comes to rest. Find the speed of C after the second impact.



- 6 Three balls, A , B and C , of masses 4 kg , 1 kg and 2 kg , respectively, are at rest in a straight line on a smooth horizontal surface, with B between A and C . A is given an initial velocity of 5 m s^{-1} towards B . When A hits B they coalesce and continue as a single object, D , until they collide with C . After this collision C has velocity 5 m s^{-1} . Work out the final velocity of D .
- 7 Jayne is performing in a show on ice. She is pushed onto the ice while sitting on a chair. The chair slides across the ice and Jayne then stands up and moves away from the chair. Jayne has speed 4 m s^{-1} when she is sitting on the chair and speed 5 m s^{-1} when she moves away from the chair. Jayne has mass 60 kg and the chair has mass 6 kg .
- Find the velocity of the chair as Jayne moves away from it.
 - What modelling assumptions have you made?
- 8 A bean bag of mass 100 g is thrown at 5 m s^{-1} at a stationary target. The bean bag sticks to the target and they move off together at 0.1 m s^{-1} . Find the mass of the target.
- 9 Mariam is moving on a sledge at 2 m s^{-1} . The combined mass of Mariam and the sledge is 40 kg . Sarah, who has mass 60 kg , runs up behind the sledge and jumps onto it. The sledge continues in the same straight line with speed 2.3 m s^{-1} .
- Find Sarah's speed just before she lands on the sledge.
 - What assumption have you made regarding Sarah's velocity?

7. Work and energy

a. Work done by force

- 6 A horse-drawn barge is pulled 40 m forwards using a rope at an angle θ to the direction of motion. The tension in the rope is 150 N. The barge is kept moving in a straight line by a contact force with the edge of the canal. Resistance forces also act on the barge.
- Determine the work done by the tension:
 - when $\theta = 10^\circ$
 - when $\theta = 20^\circ$.
 - Show that when $\theta = 20^\circ$ the tension would need to increase to 157.2 N to do the same work as in part ai. Consider the barge being pulled with a tension of 150 N with $\theta = 10^\circ$, or being pulled with a tension of 157.2 N with $\theta = 20^\circ$.
 - Explain why the frictional resistance will be greater in the second of these situations.
- 7 A box is pulled 2 m across a horizontal floor, using a rope with tension 10 N at 30° to the horizontal. The frictional resistance is 3 N. Find:
- the work done against friction
 - the work done by the tension
 - the work done by the weight
 - the work done by the normal contact force
 - the total work done by all four forces.
- 8 A crate of mass 25 kg slides 4 m down a slope that is inclined at 15° to the horizontal. Non-gravitational resistance is 5 N. Find:
- the work done against non-gravitational resistance
 - the work done against gravity
 - the work done by the normal contact force
 - the total work done by all these forces.
- 9 A crate of mass 25 kg is pushed 4 m up a slope that is inclined at 15° to the horizontal by a force of 100 N parallel to the slope. Non-gravitational resistance is 5 N. Find:
- the work done by the force of 100 N
 - the work done against non-gravitational resistance
 - the work done against gravity
 - the work done against the normal contact force
 - the total work done by all these forces.
- 10 A tile of mass 0.5 kg slides 2 m down a roof, which is inclined at 60° to the vertical. The frictional force is 1.5 N. There are no other external forces. Find:
- the work done by gravity
 - the work done against friction
 - the work done by the normal reaction force
 - the total work done by all three forces.

b. Kinetic energy

- 6 A model train of mass 1 kg is moving at 3 m s^{-1} . It accelerates uniformly for 5 s, travelling in a straight line and covering a distance of 40 m while accelerating.
- Find the speed of the train at the end of the 5 s.
 - Find the increase in kinetic energy from the start to the end of the 5 s.
- 7 A ball bearing with mass 0.03 kg is projected vertically upwards. It loses 0.735 J of kinetic energy before coming to instantaneous rest. Find the initial speed of the ball bearing.
- 8 A box of mass 30 kg slides from the top of a smooth slope to the bottom of the slope. The slope is inclined at 30° to the horizontal. The box starts from rest. At the bottom of the slope the box has gained 375 J of kinetic energy. Find the length of the slope.
- 9 A boy of mass 64 kg runs at a constant speed along a straight track. He takes 16 s to run 100 m.
- Work out his kinetic energy.
 - What difference would it make if the track was curved?
- 10 At its launch a rocket has mass 2 million kg. It accelerates from rest to 75000 m s^{-1} .
- Work out the increase in the kinetic energy.
 - Why will the calculated value be too big?
- 11 Ball *A*, of mass 2 kg, is moving in a straight line at 5 m s^{-1} . Ball *B*, of mass 4 kg, is moving in the same straight line at 2 m s^{-1} . Ball *B* is travelling directly towards ball *A*. The balls hit each other and after the impact each ball has reversed its direction of travel. The kinetic energy lost in the impact is 12.5 J.
- Show that the speed of ball *A* after the impact is $\frac{3}{10} \text{ m s}^{-1}$.
 - Find the speed of ball *B* after the impact.
- 12 Two balls, *A* and *B*, of equal mass, are travelling towards one another with velocities u_A and $-u_B$, respectively. The balls collide and their velocities after the impact are $-v_A$ and v_B , respectively. The kinetic energy after the impact is the same as the kinetic energy before the impact (i.e. a 'perfectly elastic collision'). Explain why $v_A = u_B$ and $v_B = u_A$.
- 13 Balls *X*, *Y* and *Z* lie at rest on a smooth horizontal surface, with *Y* between *X* and *Z*. Balls *X* and *Z* each have mass 2 kg and ball *Y* has mass 1 kg. Ball *X* is given a velocity of 1 m s^{-1} towards ball *Y*. Balls *X* and *Y* collide. After this collision the speed of ball *X* is 0.4 m s^{-1} in its original direction.
- Work out the loss in kinetic energy in this impact.
Ball *Y* goes on to collide with ball *Z*. After this collision the speed of ball *Y* is 0.4 m s^{-1} in the direction towards ball *X*.
 - Work out the loss in kinetic energy in this impact.
Finally, ball *Y* collides with ball *X* again. After this collision the speed of ball *Y* is twice the speed of ball *X* and the speed of ball *Z* is four times the speed of ball *Y*, with the balls all travelling in the same direction.
 - Work out the loss in kinetic energy in this impact.

c. Gravitational potential energy

- 6 A person of mass 70 kg climbs three flights of stairs to reach the third floor of a building. Each of the flights of stairs consists of 15 stairs, each of depth 18 cm. Find the increase in the potential energy of the person when they climb from the ground floor to the third floor.
- 7 A crate is pulled up a smooth slope using a rope that is parallel to the slope. The slope is inclined at an angle θ to the horizontal, where $\sin \theta = 0.28$, and the tension in the rope is 50 N. The work done by the tension is 75 J and the increase in the potential energy of the crate is 168 J. Find the mass of the crate.
- 8 A ramp rises 10 cm for every 80 cm along the sloping surface. A box of mass 50 kg slides down the ramp, starting from rest at the top of the ramp. The coefficient of friction between the ramp and the box is 0.03 and no other resistance forces act.
- Draw a diagram to show the forces acting on the box.
The box is travelling at 2 m s^{-1} when it reaches the bottom of the ramp.
 - Find the length of the ramp.
 - Find the loss in the potential energy of the box.
- 9 A boy of mass 60 kg slides down a slope that makes an angle of 45° to the horizontal. The coefficient of friction between the boy and the surface of the slope is 0.2. The boy starts from rest at the top of the slope and finishes at the end with speed $v \text{ m s}^{-1}$.
- Show that the acceleration of the boy is 5.66 m s^{-2} down the slope.
 - Work out the length of the slope in terms of v .
 - Find an expression for the loss in the boy's potential energy when he slides down the slope.
 - What modelling assumptions have you made and what effect would each of these have on your answer to part c?
- 10 The recommended slope for wheelchair ramps is 1:12. This means that the ramp rises 1 cm vertically for every 12 cm along the slope.
A person in their wheelchair, with total mass 90 kg, descend along a 1:12 wheelchair ramp that has slope length 8 m.
They start the descent with speed 2 m s^{-1} and finish with speed 4 m s^{-1} .
Work out the change in total mechanical energy (kinetic energy + potential energy) after descending the ramp.
- 11 A ball of mass 0.02 kg is projected vertically upwards through oil. The ball has initial speed $\frac{35}{12} \text{ m s}^{-1}$. The oil exerts a resistance of $0.1t^{0.5} \text{ N}$, where t s is the time from when the ball was projected.
Work out the increase in the potential energy of the ball from the start to when it comes to instantaneous rest. (Note: you will need an equation solver for this question. You will not be allowed an equation solver in the examination.)

d. Work energy principle

- 3 A crate of mass 50 kg slides down a smooth slope. At the top of the slope the crate has speed 0 m s^{-1} and at the bottom of the slope it has speed 4 m s^{-1} .
- Find:
- the increase in kinetic energy
 - the decrease in potential energy
 - the vertical height through which the crate has descended.
- 4 A boy sledges down a hill. The boy and his sledge have a combined mass of 85 kg. He starts from rest and descends through a vertical height of 3 m. Friction and air resistance are negligible.
- Find the work done by gravity.
 - Use the work–energy principle to find the boy's speed at the end of the descent.
- The boy descends the hill again, starting from rest, but this time he is joined on the sledge by his little brother, of mass 35 kg.
- Find their speed at the end of the descent.
- 5 A girl of mass 50 kg travels down a water slide. She starts at the top with a speed of 2 m s^{-1} and descends through a vertical height of 5 m.
- Assuming that there is no resistance, find her speed when she reaches the bottom of the slide.
 - The girl's actual final speed is 8 m s^{-1} because there is resistance of average value 40 N. Find the length of the water slide.
- 6 A child of mass 45 kg travels down a water chute. The child has speed 1 m s^{-1} at the top of the chute and speed 5 m s^{-1} at the bottom of the chute. The length of the water chute is 20 m and the height through which it descends is 4 m. Work out the average resistance force that acts.
- 7 A boy sits on a sledge at the top of an icy hill. He gently sets the sledge in motion. When he reaches the bottom of the hill he is moving at 10 m s^{-1} . Assuming that friction is negligible, find the height of the hill.
- 8 A girl of mass 50 kg sits on a sledge at the top of a grassy hill. She gently sets the sledge in motion. When she reaches the bottom of the hill she is moving at 9.9 m s^{-1} . The hill is 5 m high and the sledge slides 100 m down the hill.
- Work out the resistance force.
 - Comment on your answer.
- 9 A child of mass 40 kg slides down a playground slide. The child starts from rest at the top of the slide, 2 m above the ground. At the bottom of the slide its slope levels off.
- Find the child's loss of gravitational potential energy.
- There is a constant resistance of 112 N throughout.
- Find the distance the child has travelled when she comes to rest.
- The slide is inclined at an angle of 30° to the horizontal.
- Find the distance the child travels on the level part of the slide.
- 10 A car of mass 1600 kg travels 200 m along a level road. The average driving force is 2000 N and the average resistance is 800 N. The driver claims that the speed throughout the journey was less than 30 m s^{-1} . What can you say about the initial speed of the car?

e. Conservation of energy

- 4 A box slides down a smooth ramp. The height of the ramp is 20 cm and the length of the ramp is 2.5 m. The box starts from rest. What is the speed of the box when it reaches the bottom of the ramp?
- 5 A ball is launched up a smooth slope that makes an angle 30° to the horizontal. The ball travels a distance 2.5 m up the slope before coming to instantaneous rest. Find the launch speed of the ball.
- 6 In a pinball machine, ball bearings are fired up a slope inclined at 10° to the horizontal. After travelling 1.2 m the ball bearings reach a curved barrier. Find the maximum initial speed of a ball bearing if it comes to rest before getting to the curved barrier.
- 7 A boy sits on a sledge at the top of an icy hill. He gently sets the sledge in motion. When he reaches the bottom of the hill he is moving at $v \text{ m s}^{-1}$. Assuming that friction is negligible, find an expression for the height of the hill.
- 8 A diver jumps from a 10 m tall board into a swimming pool.
 - a The diver has an initial velocity of $u \text{ m s}^{-1}$ upwards. Find his speed when he hits the water.
 - b What modelling assumptions have been made?
- 9 A football is kicked from ground level with speed 15 m s^{-1} and rises to a height of 1.45 m. Assume that air resistance is negligible.
 - a Find the speed of the ball when it is 1 m above the ground.
At the top of its flight the ball is travelling horizontally.
 - b Explain why the horizontal component of the velocity is constant throughout the motion.
 - c Show that the ball was kicked at an angle of 21.0° with the horizontal.

- 4 A skydiver of mass 80 kg falls 1000 m from rest and then opens his parachute for the remaining 2000 m of his fall. Air resistance is negligible until the parachute opens. The skydiver is travelling at 5 m s^{-1} just before he hits the ground. Find the average resistance force when the skydiver is falling with the parachute open.
- 5 A tile of mass 1 kg slides 3 m down a roof inclined at 20° to the horizontal. The tile then falls 5 m under gravity to hit the ground with speed 8 m s^{-1} . Find the frictional force between the tile and the roof.
- 6 A model racing car of mass 50 g is released from rest at the top of a downward-sloping track. It travels along the track under the action of gravity. The resistance to the motion of the car is 0.05 N. The car comes to a stop on a horizontal piece of track that is 2 m lower than the top of the track. Find the distance that the car has travelled.
- 7 A diver jumps from a 10 m board above a swimming pool. The diver has an initial velocity of $u \text{ m s}^{-1}$ upwards. The horizontal component of the diver's path is negligible. A constant resistance of 0.5 N kg^{-1} acts on the diver. Find an expression for the height of the diver above the pool at the highest point of her dive.
- 8 A golf ball of mass 45.9 g is hit from a tee with speed 50 m s^{-1} . The ball lands in a pond that is 5 m lower than the tee. When the ball lands in the pond it has travelled along a curved path of length 160 m. The resistance acting on the ball has magnitude 0.3 N.
 - a Find the speed of the ball just before it hits the water.
The water immediately absorbs 8 J of energy from the ball. The ball then sinks vertically downwards to reach the bottom of the pond. The resistance acting on the ball has magnitude 3 N and the ball just comes to rest as it reaches the bottom of the pond.
 - b Find the depth of the pond.
- 9 A golf ball of mass 45.9 g is hit from a tee with speed 180 km h^{-1} . The ball rises to a height of 20 m, having travelled along a curved path of length 61.875 m. At the highest point of its path the ball is travelling at 144 km h^{-1} .
 - a Find the magnitude of the average resistance force acting on the golf ball.
The ball travels a further 105.8 m along a curved path to land on the green. The green is 4 m lower than the tee. The average resistance remains unchanged.
 - b Find the speed of the ball just before it lands on the green.
The ball is travelling vertically when it lands on the green, where it is immediately brought to rest.
 - c Show that the energy absorbed by the green is 30.3 J.
- 10 Two particles, *A* and *B*, are connected by a light inextensible string. Particle *A* has mass 2 kg and particle *B* has mass 5 kg. The string passes over a pulley and hangs vertically with particle *A* and particle *B* on each side of the pulley. The pulley, however, is not smooth and 10 J of energy is dissipated for each rotation of the pulley. The system is released from rest, and the particles reach a speed of 0.2 m s^{-1} after each moving 1.6 m.
 - a Work out how many rotations the pulley has made.
 - b If the string passes over the pulley without slipping, work out the radius of the pulley.

f. Power

- 2 The engine of a 5 tonne truck has a power output of 400 kW. The truck is travelling in a straight line on a horizontal road. The resistance to motion is 20 000 N. Find the maximum speed the truck could achieve.
- 3 The maximum power of a boat engine is 140 kW. The boat is subject to a resistance force of 10 000 N. Find the maximum speed the boat can achieve when travelling in a straight line.
- 4 A model train of mass 200 g is moving in a straight line on a level track. The train accelerates from 2 m s^{-1} to 8 m s^{-1} in 3 s. Find the average power generated by the engine.
- 5 The maximum power of a boat engine is 120 kW. The maximum speed the boat can achieve is 10 m s^{-1} . Find the resistance force acting on the boat when it is travelling at its maximum speed and the engine is working at maximum power.
- 6 A car of mass 1800 kg is being driven, at a constant 15 m s^{-1} , against a resistance of 2000 N, up a hill inclined at 10° to the horizontal. Find the rate at which the engine is working.
- 7 A car of mass 1600 kg is being driven up a hill inclined at 10° to the horizontal. The car has an initial speed of 10 m s^{-1} and a final speed of 12 m s^{-1} after 60 s. Air resistance and friction may be ignored. Find the average power generated by the engine.
- 8 A car of mass 1200 kg accelerates up a hill against a resistance of 263 N. At a certain point on the hill the road is inclined at 8° to the horizontal. The engine is working at 75 kW and the car is travelling at 25 m s^{-1} . Find the acceleration of the car at this instant.
- 9 A small van of mass 1600 kg accelerates from rest in a straight line along a horizontal road. The resistance from friction and air resistance is 2400 N throughout the motion. The engine works at a constant rate of 48 kW.
 - a Write down an expression for the driving force when the van is travelling at $v \text{ m s}^{-1}$.
 - b Write down an expression for the acceleration of the van when it is travelling at $v \text{ m s}^{-1}$.
 - c Explain why the power cannot be constant.
- 10 A powerboat of mass 500 kg travels in a straight line at its maximum velocity against a resistance of 15 N. The engine of the powerboat has a maximum power output of 750 W.
 - a Find the maximum velocity.

In different weather conditions the same powerboat has a maximum velocity of only 25 m s^{-1} .

 - b State what has changed in the model and give the new value of this quantity.
- 11 A van of mass $m \text{ kg}$ moves up a hill that is inclined at 2° to the horizontal. The engine works at a constant rate of 30 kW and the resistance (from friction and air resistance) is a constant 400 N. When the van is travelling at 20 m s^{-1} it has acceleration 0.1 m s^{-2} . Find the value of m .
- 12 Car A, of mass 1250 kg, is travelling along a straight horizontal road at speed 10 m s^{-1} . The engine works at a constant rate of 25 kW and the resistance is a constant 500 N. After 5 s the speed of the car has increased to $v \text{ m s}^{-1}$.
 - a Use the work–energy principle to find the amount of energy that is dissipated and, hence, find the distance travelled in the 5 s.
 - b Find an expression for the acceleration at time 5 s as a function of v and show that the acceleration is not constant.

Car B travels along the same road, starting with speed 10 m s^{-1} and accelerating at a constant rate for 5 s. After 5 s the two cars have the same speed and also have the same acceleration as one another.

 - c Show that v must satisfy the equation $v^2 - 8v = 100$ and, hence, find the speed of the cars at the end of the 5 s.